

■ NSum

`NSum[f, {i, imin, imax}]` gives a numerical approximation to the sum $\sum_{i=imin}^{imax} f$.

`NSum[f, {i, imin, imax, di}]` uses a step di in the sum.

`NSum` can be used for sums with both finite and infinite limits. ■ The following options can be given:

<code>AccuracyGoal</code>	6	number of digits of final accuracy to try and get
<code>ExtraTerms</code>	11	maximum number of terms to use in extrapolation
<code>Method</code>	<code>Automatic</code>	method to use: <code>Integrate</code> or <code>Fit</code>
<code>Terms</code>	15	number of terms to use before extrapolation
<code>WorkingPrecision</code>	<code>Precision[1.]</code>	the number of digits used in internal computations

■ `NSum` uses either the Euler-Maclaurin (`Integrate`) or Wynn epsilon (`Fit`) method. ■ You should realize that with sufficiently pathological summands, the algorithms used by `NSum` can give wrong answers. In most cases, you can test the answer by looking at its sensitivity to changes in the setting of options for `NSum`. ■ `N[Sum[...]]` calls `NSum`. ■ See page 474.